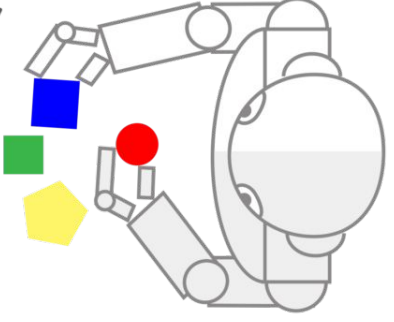




Towards Robust Prehensile Manipulation in Open-Ended Environments

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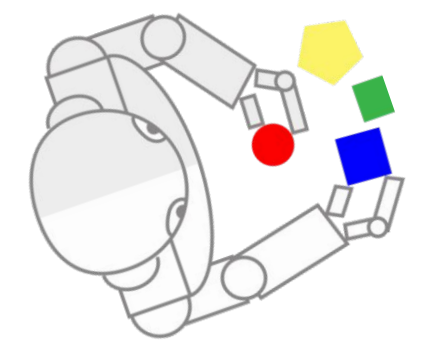
Abstract : This work proposes the use of Quality-Diversity (QD) algorithms to solve robotic prehensile manipulation tasks in open-ended environments.

Introduction

Challenge of Learning to Grasp in Robotics

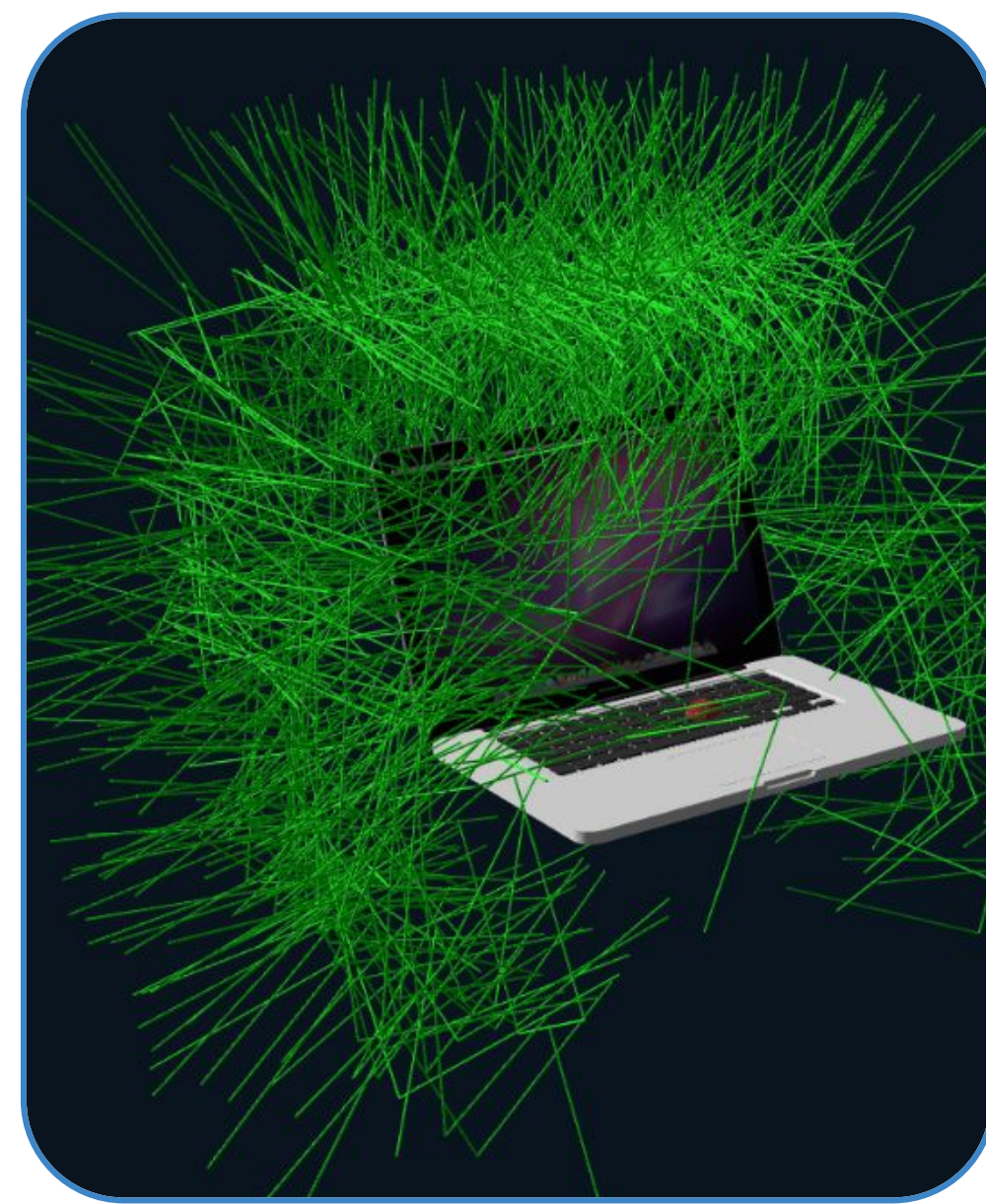
- Grasping $\Rightarrow$ sparse reward task
- Data-Driven methods require large demonstration dataset

How to automatically generate prehensile manipulation demonstrations in a sparse reward environment ?



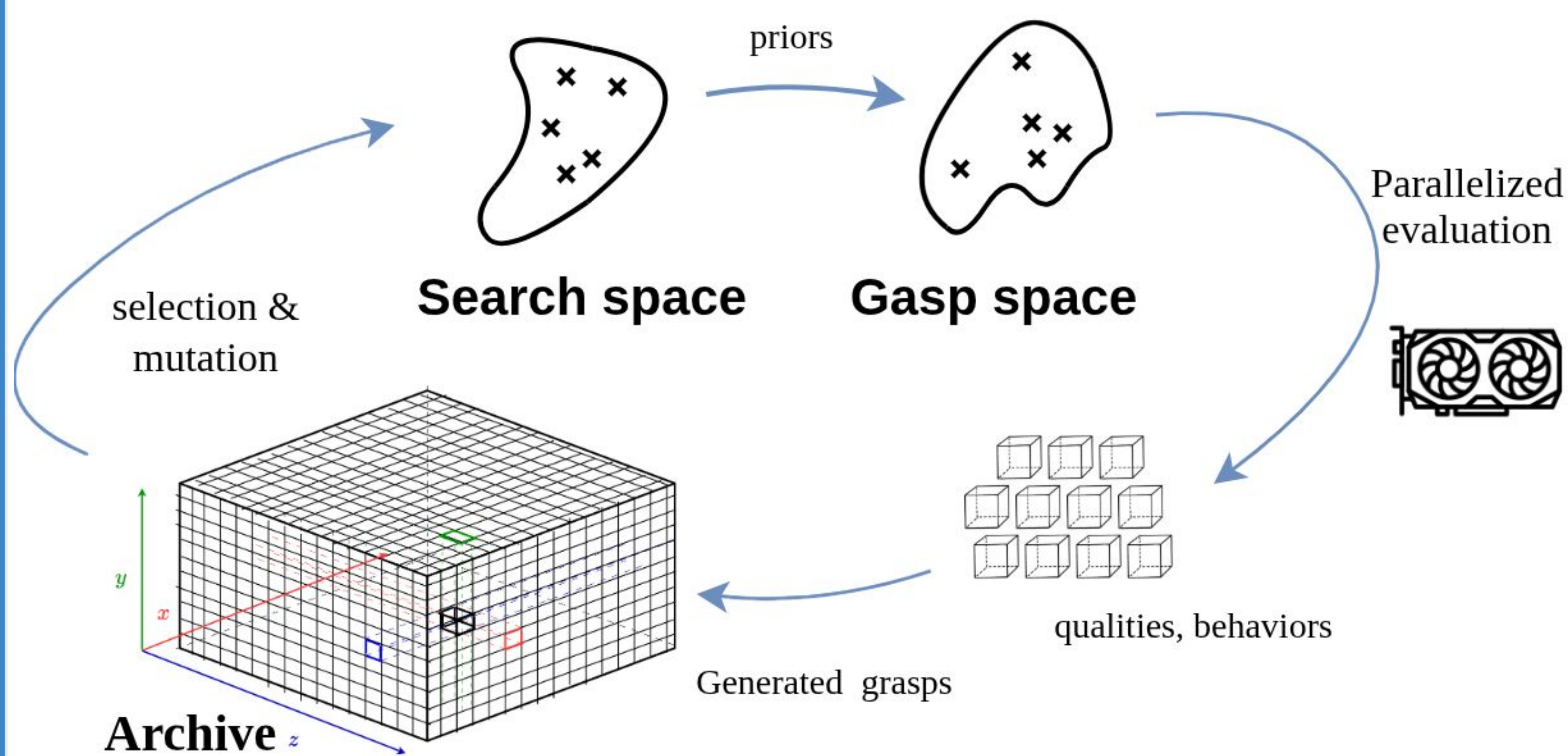
Problem Framing

- Dataset of object-centric grasp demonstrations $g \in SE(3) \times \mathbb{R}^n$
- One demonstration = one 6-DoF end-effector grasp in the object reference frame [1][2]



Generating grasps with Quality-Diversity (QD)

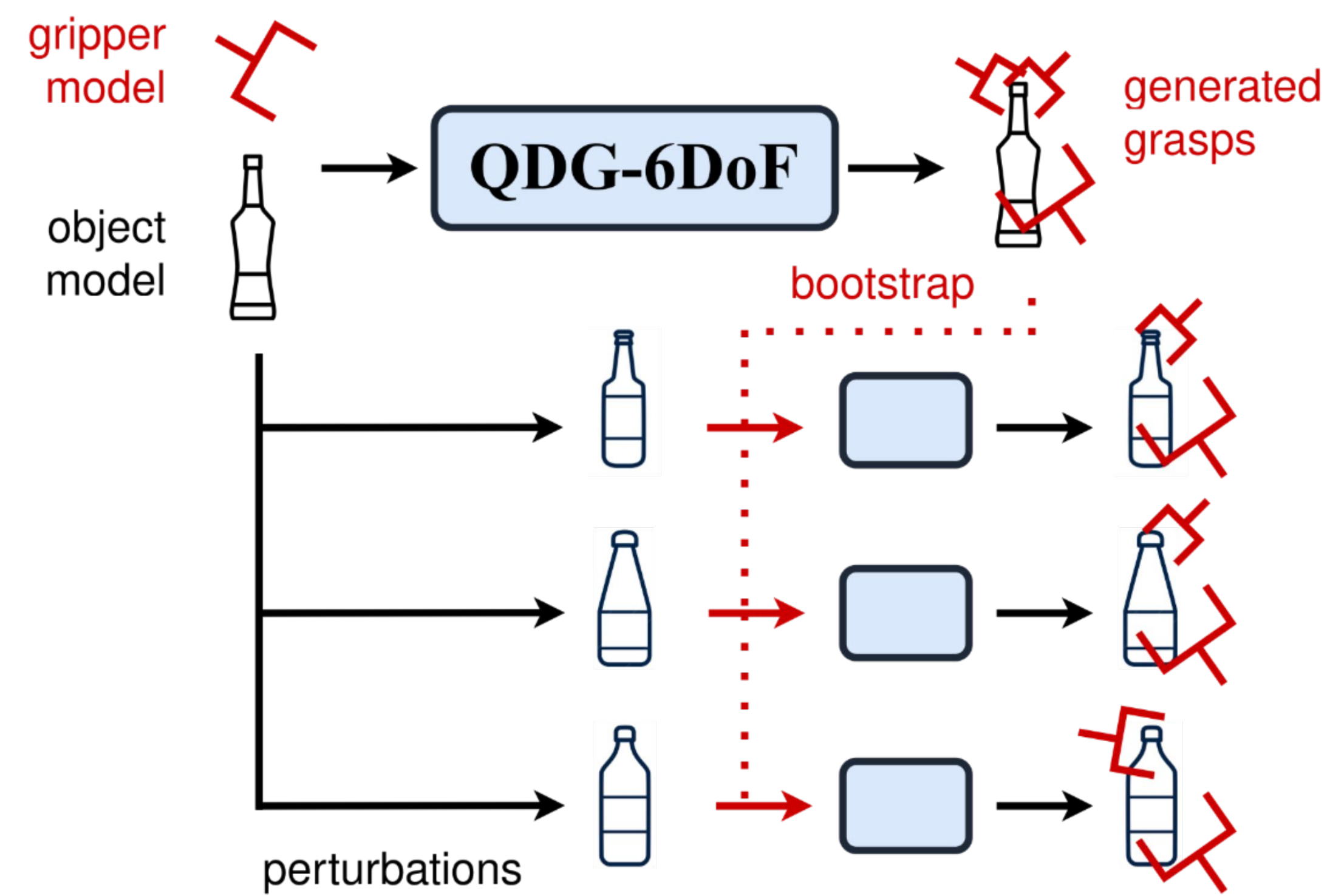
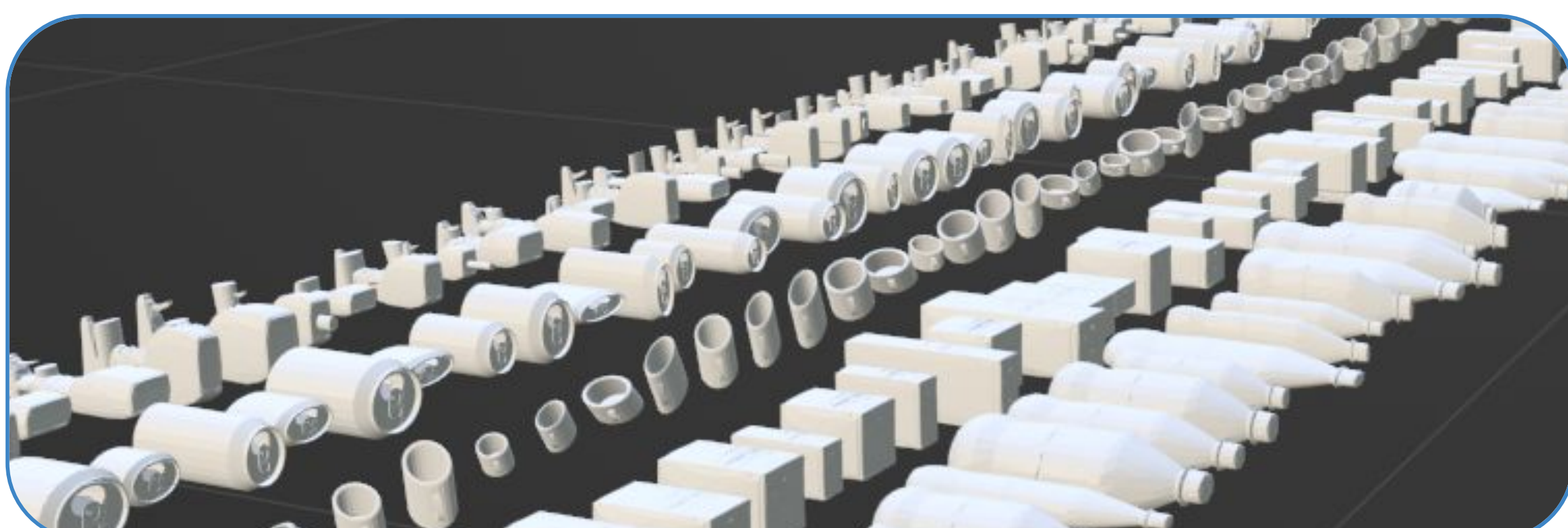
- $\left\{ \begin{array}{l} \forall b \in \mathcal{B}_{reach}, \exists \theta \in A, d_{\mathcal{B}}(\phi_{\mathcal{B}}(\theta), b) < \epsilon \text{ (diverse solutions)} \\ \forall \theta' \in A, \theta' = \operatorname{argmax}_{\theta \in N(b_{\theta'})} f(\theta) \text{ (best-performing solutions)} \end{array} \right.$
- Efficient exploration of sparse reward environments
- Evolutionary algorithm $\Rightarrow$ selection-mutation loop $\Rightarrow$ MAP-Elite



Methods

Using Bootstrap to Scale Up the Production of Synthetic Grasping Datasets

- Run QD optimisation on a given object A_s^b
- Use A_s^b to bootstrap the optimisation on similar objects [3]



- Core initial object dataset $P, n = 29\,651$ objects
 $\mathcal{P}_i = \{p_j \in \mathbb{R}^3 \mid p_j = (x_j, y_j, z_j), j = 1, 2, \dots, m_i\}$
- Data augmentation = point cloud deformation

$$\mathcal{P}_i^a = \{\varphi_{D_a}(p_j) = D_a \cdot p_j \mid p_j \in \mathcal{P}_i\}$$

where $\varphi_{D_a} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$\begin{pmatrix} x_j \\ y_j \\ z_j \end{pmatrix} \mapsto \begin{pmatrix} \alpha_1 & 0 & 0 \\ 0 & \alpha_2 & 0 \\ 0 & 0 & \alpha_3 \end{pmatrix} \begin{pmatrix} x_j \\ y_j \\ z_j \end{pmatrix}$$

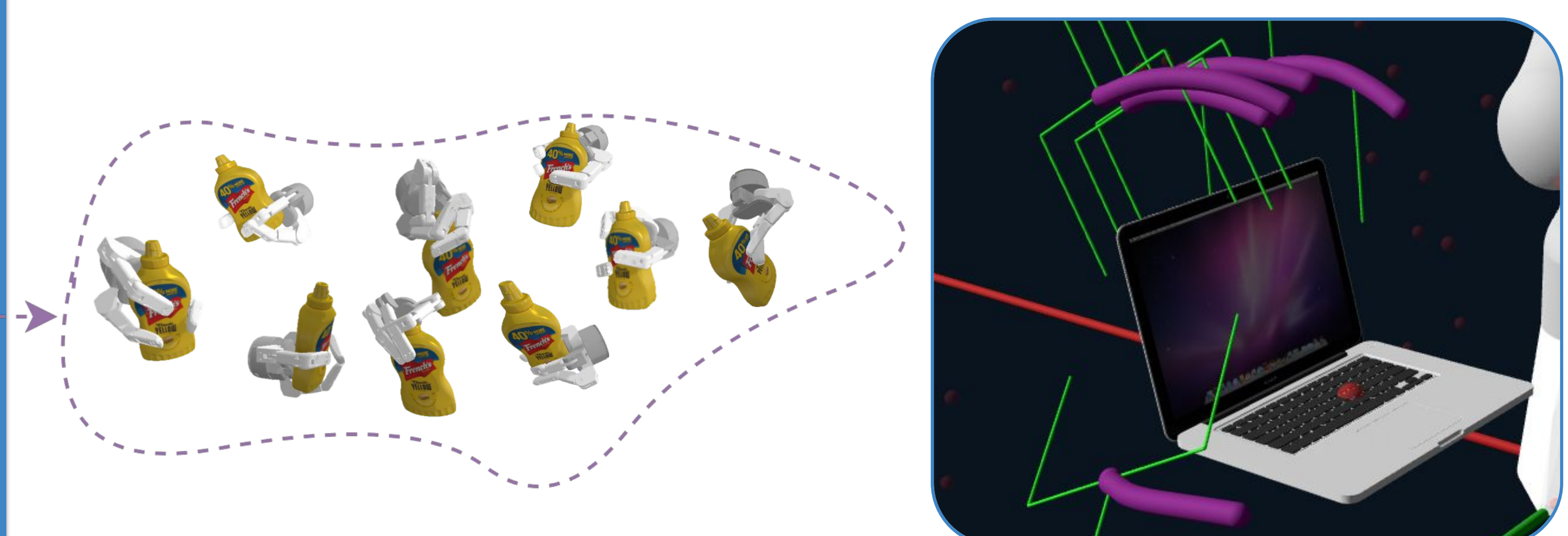
- Final dataset size = **40 353 objects**

Using bootstrap to generate manipulation trajectories

- Each stable grasp is the starting point of a manipulation trajectory



Results



- QDG-6DoF : compatible with various robotic platforms
- QDGset :
 - 62 million** grasps across 5 000 distinct objects
 - serves as a rich source of initial configurations for **generating diverse manipulation trajectories**

References

- [1] Huber, J., Hélienon, F., Kappel, M., Chelly, E., Khoramshahi, M., Amar, F. B. & Doncieux, S. (2024). Speeding up 6-DoF Grasp Sampling with Quality-Diversity. (IROS'24)
- [2] Huber, J., Hélienon, F., Kappel, M., de Loyola Páez-Ubieta, I. L., Puente, S. T., Gil, P., Ben Amar, F. & Doncieux, S. (2024). QDGset: A Large-Scale Grasping Dataset Generated with Quality-Diversity. (ICRA'25)